Capstone Meeting Summary

Adaptive Writing Tutor | October 7, 2026

The meeting clarified the January prototype scope, options for expanding training data, and the intended role of the multi-armed bandit. The current SRS requirements were generally considered appropriate, with some adjustments needed.

1. January prototype and hosting

The January pilot will primarily focus on **collecting useful data**, rather than running a complete experiment. The minimum prototype should provide a writing space, basic AI feedback through an SLM, and data logging. **The multi-armed bandit does not need to be ready by January.**

The prototype is expected to run on McMaster servers, while development can continue locally. Hosting arrangements are in progress. McMaster SSO will require coordination with UTS and may involve a security review.

2. Training data and SLM fine-tuning

The preferred dataset contains fewer than 600 examples and includes student writing, instructor feedback, rubrics, and scores. Existing data is intended for initial prototyping; **no minimum dataset size was agreed upon.**

The team should explore adapting other available datasets instead of waiting until January. Angela accepted using an LLM to generate or adapt feedback for existing writing samples, provided that the process follows specific rubrics. Amber also highlighted concerns about the quality of publicly available synthetic writing datasets.

Amber will share a UTM paper containing a rubric for common issues in CS writing. Angela will contact the researchers about access to an anonymized dataset. Data from an earlier voice-versus-text study may also be reusable.

3. January data collection

Angela's first-year discrete mathematics course has approximately 200 students and may provide opportunities for writing activities. The amount and type of data collected will depend on the team's needs, activity design, and availability of human graders.

Possible background information includes technical writing experience, motivation, and attitudes toward receiving feedback. The final questionnaire and collection scope remain open.

4. SRS requirements and experimental groups

Research consent and withdrawal can be handled separately from the application. The research team would exclude data from students who do not consent before using it for research.

Instructors should manage experimental groups and student assignments to conditions. Students should not create or choose their own experimental groups. Student-created groups or collaboration features may remain optional future additions.

Students can choose whether to request AI feedback. This choice is separate from research participation. The current SRS can retain the existing dataset plan while alternative data sources are investigated.

5. Multi-armed bandit scope

The discussion identified two related goals: adapting feedback for students and providing a broader framework for testing different tool features.

The agreed starting point is to **test different ways of giving feedback and optimize feedback selection for students**. Grammar, clarity, and organization were discussed as possible dimensions to measure, but the exact arms, reward signals, and evaluation metrics still need to be defined.

A more configurable framework, allowing instructors to choose experimental conditions and outcome variables, can be explored later. Comparing feedback presentation styles or voice versus text feedback were examples of future possibilities, not immediate requirements.

6. Next steps

- **Team:** Share the SRS link, revise the grouping and research-related requirements, and investigate adapting additional datasets.

- **Angela and Amber:** Review the SRS and provide further comments.

- **Amber:** Share the UTM rubric paper.

- **Angela:** Investigate additional dataset access and coordinate potential January course activities.

- **Team and supervisors:** Refine the initial bandit experiment and its evaluation metrics.

- **Team:** Decide whether to meet during Reading Week and notify the supervisors in advance.